

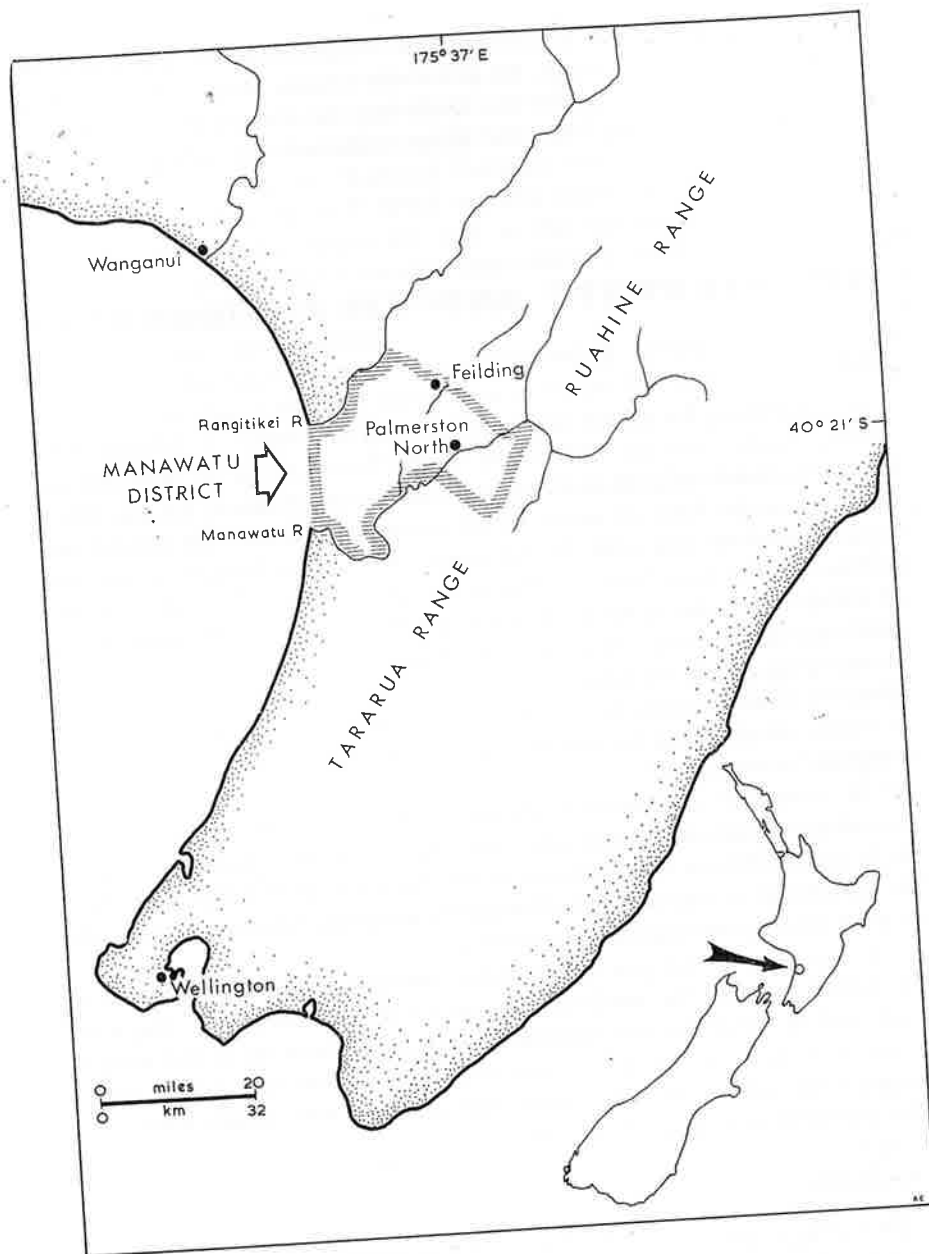


Figure 1 Location of the Manawatu district, New Zealand.

The coast is sandy throughout its entire 19 km. For centuries sand has been deposited on the beach and driven inland by the wind. Behind the coastal strip of moving sand (first dune phase) are extensive plains and long, low parabolic dunes which have been stable for 5 centuries or more (second dune phase). Eastwards from this another set of plains and dunes reaching 19 km from the coast represents a period of sand movement which ceased about 30, or possibly 50, centuries ago (third dune phase).

River flats built up by three rivers, the Rangitikei, Manawatu, and Oroua, cover about one half of the Manawatu lowlands. The Rangitikei River rises in the centre of the North Island. In the meandering lower reaches it has formed river flats but these make up only a minor proportion of the Manawatu land surface. Far greater in importance is the Manawatu River which drains 594 600 ha on both sides of parts of the Ruahine and Tararua Ranges and winds its lower course of nearly 80 km through the lowlands less than 60 m above sea level. The river has a shingle bed to Palmerston North and a short distance beyond, but further downstream it flows sluggishly through fine silty alluvium in a meandering course. A tributary, the Oroua River, is a much smaller stream but has made a major contribution to the building of the plains. Peat swamps fill large expanses of low-lying country, overlying relics of forest in many places.

Extensive terraces occur in three regions—between the Manawatu River and the Tararua Range, north-east from Palmerston North, and between the Oroua and Rangitikei Rivers. The terraces east of the Manawatu River are narrow at the gorge but several kilometres wide in the south-west. There is a low terrace bordering the river, an intermediate one some metres above it, and a much more extensive high terrace butting against the greywacke (*figures 2 and 3*). This high terrace, the oldest, is composed of sandstone and gravel and is deeply dissected by steep-sided valleys separated by broad, flattish interfluvies.

From Palmerston North broad terraces rise steadily towards the north-east. The topography is gently rolling and little dissected but with prominent scarps near the Manawatu River.

The terraces between the Rangitikei and Oroua Rivers are well defined on their margins but warping of the land surface has raised the centre into a long rounded ridge on which lie the trig stations Mt Stewart 131 m, Mt Lees 138 m, and Mt Biggs 145 m.

The intermediate and the upper terraces are deeply mantled with loess lifted from the Manawatu and Rangitikei Rivers during the last period of glaciation. The deposits are shallower away from the rivers.

The greywacke hills mark the eastern margin of the Manawatu. The Tararua Range forms part of the backbone of the southern part of the North

Island. In the north it descends in a long gentle slope from an altitude of more than 1 550 m in the main part of the range to little more than 300 m at the Manawatu Gorge. The portion in the Manawatu district is all below 767 m. The level to rolling country on the crest of the range represents an old peneplain surface. Loess and greywacke erosion debris and volcanic ash lie quite deep in some places. This easy topography is broken by deeply entrenched streams and by the western flank of the greywacke which drops fairly steeply to the terraces on its margin.

In this handbook four terms have been adopted in reference to these physiographic units: *Lowlands* refers to sand country and the river flats; *Terraces* embraces the three areas of terraces; *Downlands* refers to the rolling terrace country in the Mt Stewart - Mt Biggs region; *Hill country* (or hills) is used in reference to the greywacke hills only.

Soils

Soils of the Manawatu are grouped broadly into yellow-brown sands and associated gley and organic soils on the coast; recent and organic soils on the river flats; yellow-grey earths on the terraces and downlands; and yellow-brown earths and related steepland soils on the greywacke hills.

Sand country soils vary widely, particularly with age and moisture level. Dune sand which has recently become stabilised differs little from the mobile sand. On ageing a darker topsoil develops as organic matter is incorporated and the subsoil becomes more yellowish brown; further ageing deepens and darkens the topsoil and extends the subsoil. Soils on most parts of the sand plains are influenced by the position of the water table. Where the water table is very high the typical blue-grey subsoil of a gley develops. Peat forms where wet conditions favour the accumulation of plant remains. Drier sand develops a black topsoil and grey subsoil, often with mottles or concretions of iron. Topsoils in the non-peaty places are mostly dark brown to black, a little more than 5 cm deep near the coast but up to 30 cm deep in the older soils. They also become more loamy with age.

Soils of the river flats vary widely depending mainly on the nature of the alluvium, the frequency of flooding, and the natural drainage. The coarsest alluvium has been deposited near the streams. These soils are the most free-draining and the youngest. They show little differentiation into topsoil and subsoil. A little further from the river lies a set of soils ranging from sandy loams to silt loams, and from free-draining to poorly drained. In the very low-lying backwater swamps there are peaty soils which have been built by the slowly decomposing remains of swamp plants.

The soils of the terraces are formed from loess and are characterised by greyish brown loamy topsoils overlying olive to olive grey heavy textured and



Figure 2 The junction of the terraces and the greywacke hills at the mouth of the Kahuterawa valley. Greens Rd is in the foreground. The Linton district and the lowlands are in the distance. 1967

compact subsoils with yellowish brown mottles. All these soils are fairly acid and poorly drained. The soils on the low terrace and the fans are more gravelly. In the region of Forest Hill Rd there is a very infertile soil on sites previously occupied by beech forest.

The soils of the downlands are essentially the same as those of the terraces except that the loess is shallower over the underlying sandstone.

Soils of the hill country have formed either on greywacke on the steeper parts or on greywacke with the addition of loess and volcanic ash on the gentler topography. The steepland soils derived directly from greywacke are shallow and stony. Deeper and more friable soils occur on the easier slopes but both are moderately or strongly leached and fairly acid. This is partly a reflection of the higher rainfall in the hill country.

The geology and soils of the Manawatu have been treated very briefly here. For further details see Cowie and Smith (1958), Cowie 1963; 1964a; 1964b)

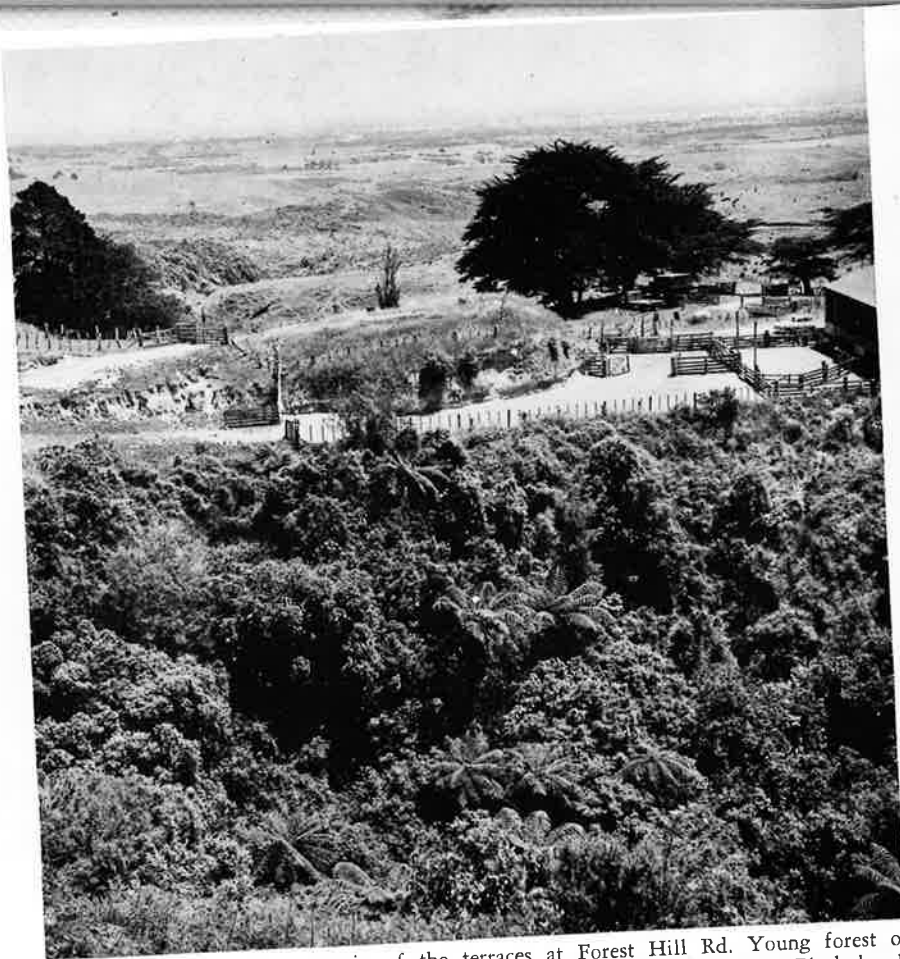


Figure 3 The eastern margin of the terraces at Forest Hill Rd. Young forest of native plants fills the gully of the deeply entrenched Te Taupiri Stream. Black beech forest formerly occupied about 5 sq. km in this district. 1960

and Cowie *et al.* (1967). At the time of writing this handbook the survey of soils of the Kairanga County by Cowie is in press.

Climate

The main features of the Manawatu climate are the moderate temperatures, moderate rainfall, moderate hours of sunshine, and the considerable amount of wind.

At Palmerston North the January mean temperature is 16.6°C and the June mean 8.1°C. There are about 60 ground frosts a year and snow lies briefly on the hills each winter.

Annual rainfall increases from a little more than 750 mm on the coast to about 1000 mm at Feilding and Palmerston North to around 2000 mm on the Tararua Range. There are on average 173 rain days per year at Palmerston

North, fewer on the coast and more in the hills. Rainfall is fairly well distributed throughout the year and severe droughts occur infrequently.

The Manawatu is a fairly cloudy district with only 1 814 sunshine hours recorded annually at Palmerston North. Cloud frequently hangs over the Tararua Range and sunshine hours there are probably not more than 1 400. The coast is much less cloudy and has over 2 000 sunshine hours annually.

Very strong and persistent winds are a feature of the Manawatu. Most of the winds are from the west and north-west, occurring on nearly half of the 320 wind days at Palmerston North each year. Gales are fairly frequent and may occur at any month of the year with lower frequencies in winter months when there are 4 or 5 days each month when gusts exceed 64 km per hour. At other times of the year gusts reach this speed on 6 to 8 days each month.

HISTORY OF THE DISTRICT

The aspects of history considered in this handbook are those relating fairly directly to the plant life. These are principally the developments associated with the changing landscape, the change from a wholly indigenous vegetation to one conspicuously alien, and the role of native plants in the economy of the district.

In Maori times the Manawatu was sparsely populated in comparison with many other North Island districts. The unstable sandy coast offered no suitable sites for habitation and inland sand country provided little food or security from enemies. The large tracts of forest precluded occupation of much of the remainder of the district. Maori settlements were established mainly on the margins of the rivers, where there was soil that could be cultivated and the river served as the main inland means of communication. Nearby forest provided refuge and some food. The clearings occupied by the pas and kaingas may have been natural clearings but I believe forest was removed from selected areas of ground above flood level.

A record of some of the events in the district before 1850 is to be found in Buick (1903).

Settlement

European settlement of the Manawatu began relatively late. Some traders knew the lower reaches of the Manawatu River in the 1830s but it was not until the New Zealand Company began making surveys and reports between 1840 and 1843 that much European interest was aroused. In 1840 E. J. Wakefield briefly visited the Foxton area and later travelled on foot from Parewanui, near the Rangitikei River, to the Oroua River. Missionaries,

of sand country drew many visitors away from the easy routes along the beaches.

(3) The region lacks prominent features such as lakes, bluffs, or outstanding mountain peaks likely to attract the special attention of plant hunters and explorers.

William Colenso, an inveterate traveller, saw most of the North Island but did not visit the area under study. He was very active in southern Hawke's Bay around the headwaters of the Manawatu River and crossed the Ruahine Range into the Pohangina, a tributary of the Manawatu River. Richard Taylor, a missionary working in the Wanganui, South Taranaki, and Rangitikei districts from 1843 to 1873, passed through the Manawatu many times and recorded some comments about the vegetation in his diaries.

In 1874 J. Buchanan published "Notes on the flora of the Province of Wellington, with a list of plants collected therein" (Buchanan 1874). He limited his area to the region south of a line running from the Wanganui River to Castlepoint. Unfortunately, his only reference to Manawatu is a locality citation for "*Plantago brownii*". From this and other writings it appears that the Manawatu held little particular interest for him.

Another wide-ranging botanist, T. F. Cheeseman, also seems to have passed by the Manawatu, as the region does not figure in any of his extensive writings. D. Petrie visited the Manawatu Gorge in 1889 (Hamlin 1958a) but apparently left no botanical records. It seems that B. C. Aston can be credited with the first description of a plant community in the Manawatu. In a series of three articles on the soils of the region he commented on Round Bush (Aston 1920a) and listed the significant species in the Esplanade, Palmerston North (Aston 1920b). His earlier paper (Aston 1910) contained only one minor reference to the Manawatu.

Dr L. Cockayne visited the Manawatu on many occasions but references to its vegetation and flora are few. A very general account of some of the botanical features (Cockayne 1921) embraces the Horowhenua district as well. No specific mention of the region is made in his sand country report (Cockayne 1911) but I have the impression that the coastal strip just south of the Manawatu River strongly influenced his classification of the sand country communities. In his *Vegetation of New Zealand* (Cockayne 1928) references are more to the geographical features than to the botany. However, his figure 30 is of some interest as it is one of the first published photographs taken specifically to show an aspect of the vegetation of the Manawatu. The photographer, H. H. Allan, a teacher at Feilding Agricultural High School from 1923 to 1927, became a very prominent botanist and was responsible for recording many aspects of the botany of the district. In 1924 appeared "Forest remnants in the neighbourhood of Feilding" (Allan 1924c) and later an

account of the introduction of *Spartina* into New Zealand and its establishment at Foxton (Allan 1924a). He wrote, "In 1913 a number of plants of *Spartina* were obtained by Mr Dalrymple from Southampton Water and planted out in small clumps on the Foxton mud-flats in the tidal estuary of the Manawatu River. The clumps were considered to consist of specimens of both *S. stricta* and *S. townsendii*, but unfortunately it is not certain that the latter has established itself". A brief but useful description of the salt swamp was given in the same paper. In an article a few years later (Allan 1930) he outlined the confusion over the naming of the species and concluded that "*Spartina townsendii* has definitely been established in New Zealand at Foxton". He was also able to produce graphs showing the rate of spread of patches from 1913 to 1930.

In collaboration with L. Cockayne he described and named two new plants from the Manawatu. One was the lianoid *Fuchsia perscandens* from Kitchener Park (Cockayne and Allan 1927, page 53), the other was *Libertia peregrinans* from "sand hollows near Foxton" (Cockayne and Allan 1927, page 56). In the same paper mention was made of the similarity of the *Teuclidium parviflorum* growing between Pohangina and Feilding to the one described by T. F. Cheeseman as var. *luxurians* (Cockayne and Allan 1927, page 57).

A significant paper by Allan and two overseas botanists as co-authors, (Zahlbruckner *et al.* 1928) described several new species of lichens and gave some detail of the structure and composition of Kitchener Park where they were collected. About this time Allan was taking an interest in wild hybrids in the New Zealand flora. His records included many species, both native and exotic observed in the Manawatu (Allan 1929, Cockayne and Allan 1934, and a series of papers in *Genetica* volumes 7, 8, 9, and 11).

In 1928 Allan became systematic botanist with the Plant Research Station in Palmerston North. This position permitted full time study of the flora and his interests spread to a wider geographical area. His residence in the Manawatu ended in 1937 when he moved to Wellington and became the first director of Botany Division of the Department of Scientific and Industrial Research.

In 1928 Allan took onto his staff V. D. Zotov, one of his former pupils at Feilding Agricultural High School whom he described as "my keen-eyed assistant". Their joint work included the production of an artificial hybrid between *Phormium cookianum* and *P. tenax* using local plants (Allan and Zotov 1937). Zotov collaborated with G. V. Wild to give an account of interesting floral features of *Coprosma* species (Wild and Zotov 1930).

Zotov's greatest contribution to the botany of the region was his part in the joint paper on the plant life of the Tararua mountains (Zotov, *et al.* 1939). Their northern area covers part of the region which is the subject of this

handbook. Particular reference was made to the Manawatu Gorge and there is a striking photograph of supplejack in the Tiritea valley, a place that is well known today for its entanglements which have given rise to several uncomplimentary names for certain gullies in the catchment.

A. J. Healy continued the line of botanists from Feilding Agricultural High School and added his knowledge of the adventive flora, in particular, to annals of the district. Some of these records are mentioned in this handbook.

About the time that the Plant Research Station was set up Massey Agricultural College (now Massey University) came into being. Dr J. S. Yeates, its first lecturer in botany, played a very active part in the selection of strains of *Phormium*. Student theses added to the understanding of the wild or semi-wild plants of the region. D. M. E. Merry (1934) made a study of the *Juncus* species of the district; D. A. Grant (1966; 1967) studied the establishment of manuka and its regional morphological variations; and P. J. Rumball (1966) detailed the patterns of distribution of pasture-inhabiting plants in relation to microtopographic features of grazed pastures. A portion of this was later published (Rumball and Esler 1968).

In 1957 J. A. Carnahan submitted a thesis entitled "A study of ecological interaction between introduced and indigenous plant species in the Manawatu district, North Island, New Zealand" (Carnahan 1957a). His field work covered part of the Manawatu as defined here and extended up the Pohangina valley on the western side of the Ruahine Range. Some of the information was later published in a brief paper (Carnahan 1961). He also wrote a paper on the Manawatu sand country (Carnahan 1957b).

While at Massey University I published a note on *Cordyline australis* × *banksii* hybrids (Esler 1961), accounts of some lowland forest remnants (Esler 1962a), the vegetation of the Tiritea catchment (Esler 1963), the vegetation of the Manawatu (Esler 1964), and a description of Round Bush near Foxton with R. M. Greenwood (Esler and Greenwood 1968). An unpublished report with R. M. Greenwood (Esler and Greenwood 1967) to the Feilding Borough Council outlined measures for restoring Kitchener Park and gave an account of the vegetation including species lists indicating some of the changes in the flora since 1928.

Publications of mine since joining DSIR in 1969 include an account of the flowering and fruiting of nikau palms in the Kahuterawa valley near Palmerston North (Esler 1969a), an outline of the patterns of distribution of plants on sand plains in relation to depth of the water table (Esler 1969b) followed by an assessment of the role of plants in shaping the topography of the sand country (Esler 1970). An unpublished two-volume illustrated report (Esler 1969c) detailed the dynamics of the vegetation of the Tiritea reserve and documented botanical and other features of numerous clearings.

An ecological survey of the Moutoa Estate (sometimes wrongly referred to as Paiaka which was an early settlement on the other side of the river) by Botany Division, DSIR (Poole and Boyce 1949) aimed to provide plans for the rational management of the *Phormium* crop. The vegetation of the whole area was mapped on a scale of 1:3 960 (5 chains to an inch) and particular attention was paid to the ecology of *Phormium*, including determination of its place in the gamut of habitats in time and space that are a feature of wetlands. A reconstruction of the primitive vegetation is presented in this paper and more fully in another (Poole 1946). Of particular interest is a list of 252 plant species encountered on the Moutoa Estate from 1940 onwards.

Palmerston North has been a centre for plant research for more than 40 years. This activity began in 1928 with the establishment of the Plant Research Station, from which evolved several divisions of the DSIR. Grasslands, Applied Biochemistry, and Plant Physiology Divisions are based in Palmerston North. The Ministry of Agriculture and Fisheries' seed testing station and a field research unit are also located there. As most of the work of these organisations has been directed towards economic plants and is mainly on a national scale it is not reviewed here.

THE VEGETATION: PAST AND PRESENT

Reconstructing the vegetation of the past

There appears to be no botanical account of any small portion of the Manawatu vegetation before 1920. This gap in our knowledge of the early vegetation cannot be completely filled by the general descriptions of casual observers a century ago and interpretation of the present-day relics of natural vegetation. Much information is lost for ever but it is worth recording what is available at present. This study was prompted by the realisation that there still remained sources of information that would have passed into oblivion in another decade. In the 12 years during which I have been making observations much of this evidence has gone. Opossums have destroyed the last of the northern rata-kamahi forest, and bulldozers have torn into reserves which would have been expected to carry some of the story of their past. Fortunately, some vegetation is sufficiently well preserved to yield reliable information.

The reconstruction of the primitive vegetation which follows is based on written evidence, the recollections of men who knew the vegetation in its less modified state, evidence that remains in the field, and upon experience which enables rational interpretation. All these lines of approach suffer from

the weakness of subjectivity. Written and verbal accounts cannot always be verified and must be accepted where they seem reasonable. The structure and composition of the vegetation can be treated objectively but interpretation must be based on experience. Where statements are based on my conjectures this is clearly indicated lest the reader accepts thoughts that might later prove to be groundless.

Presenting an account of the early plant cover of the Manawatu is fraught with difficulties. In the first place there is no written record of the forest, swamps, shrublands, and dune vegetation which covered most of the district. Only passing comments or general descriptions of the scenery have been handed down to us. This places heavy reliance on the interpretation of the remaining vegetation to give a picture of the past. Such interpretation must take into account the changes of more than a century of European occupation. In some instances pieces of forest have been destroyed and there is no evidence of their existence other than records on early plans or decaying logs in the soil. Identification of plant remains gives some indication of the vegetation but the picture is heavily biased towards the more durable species.

Selective removal of certain species for timber and fence posts makes many forest remnants quite unlike the forests from which they were derived. Selective elimination of some species by deer, opossums, and domestic animals has had the same effect. The elimination of one or several species of plants often promotes an abundance of some others. Species best able to tolerate the present conditions tend to predominate.

Further difficulty is introduced by the uncertainty of events before European occupation of the Manawatu. At least some of the vegetation at the time of settlement was not in a truly primitive state. Whether the nature of the sand country vegetation, for example, was the result of hundreds of years of natural sorting and adjustment of plants or a response to early nineteenth century burning will never be known. It seems that a large area of forest in the Mt Stewart region, in a climate marginal for forest, had been cleared by Maori fires. It is almost certain too, that some of the non-swampy forest clearings away from the rivers were man-made.

Information from maps and plans

The vegetation features shown on the map in this handbook are from survey plans drawn between 1859 and 1881. As intensive farming had not then started, the distribution of plant communities fairly accurately portrays the situation before modification by settlement. All the information shown is from plans held by the District Office of Lands and Survey Department, Wellington. These are listed at the back of this handbook and referred to in the text by the surveyor's name and the plan number.

The accuracy of the mapping by the surveyors varies with the scale of the map or plan. J. T. Stewart's two-inches-to-the-mile (1 : 31 680) survey along the rivers (Stewart S.O. 10603) can be expected to be the least accurate. On a large area west of the Oroua River land was surveyed into quite small blocks for settlement. Here a scale of 1:7 920 (10 chains to the inch) was used and the accuracy of vegetation boundaries is high.

In some cases there are several plans for the same area. Generally each confirmed the accuracy of the others. Some discrepancies can be attributed to the time interval between the surveys but mostly to the surveyor's subjective estimate of where the vegetation boundary lay. Where swamp merges through shrubland into forest, for example, there is room for opinion in deciding where the line between swamp and forest should be drawn. It is also necessary to remember that the prime purpose of a survey was not to produce a vegetation map. We can be thankful that so much more information of secondary importance to the surveyors has been recorded. It is fortunate that many surveyors were acquainted with plants and added their comments to the plans.

The forest margins on all plans seem to have been fairly accurately indicated. However, the swamps, by their very nature, were more difficult to define. In some places continuous vegetation boundaries have not been marked but present-day physiographic boundaries are a guide to where some would have been. Very little shrubland was shown and many small swampy areas on the lowlands have not been mapped at all.

To show the positions of vegetation boundaries correctly on the map it has been necessary to redraw the river courses and place the coastal lagoons in the positions they occupied a century ago.

Forest covered half of the Manawatu. In the east it crossed the Tararua Range and extended with few breaks over the hills, terraces and lowlands to the Oroua River. Beyond there it reached from the Oroua River through Rongotea to the Rangitikei River (*map 1*). This forested area was bounded in the north by a line running from Highden towards Ohakea aerodrome, and in the south by a line from Poutu Rd through Glen Oroua to the Oroua River.

The terrace country west of Feilding was almost devoid of forest. Patches occurred along the Mangaone West Stream towards Mt Biggs and there were a number of other small patches in gullies. A finger of forest extended through Feilding towards Kitchener Park.

On the sand country patches of forest covered 2 128 ha on parts of the plains and minor areas of dunes. Apart from Round Bush and patches of forest in the Foxton borough, these were more than 8 km from the coast and were most extensive towards the eastern margin of the sand country.